ICE MAKE

Trusted Cooling Partner

Product Brochure

Cold Room-Storage



Cold Room / Storage | Industrial Refrigeration | Commercial Refrigeration | Transport Refrigeration | Ammonia Refrigeration

ICE MAKE

Trusted Cooling Partner







Ice Make is a strong player in the refrigeration arena. Ice Make provides the Industrial & Commercial Refrigeration solutions for every customer's individual needs, since 1993. This Cold Room and Storage vertical includes a product range of Cold Storage, Solar Cold Room with TES technology, Glass Door Display Chiller, Blast Freezer and Ripening Chamber along with highly efficient refrigeration systems and storage solutions. Ice Make has the infrastructural capability to produce the highly durable thermal insulated PUF panel with thickness of 60 mm to 150 mm and highly reliable refrigeration systems as per the customer's need, these products are designed to reduce the deterioration of storage foods during the freezing process, maintains food quality including flavour, texture, colour, aroma and nutrients for various industries such as Dairy, Ice Cream, Food Processing, Horticulture/Floriculture, Fruit Ripening, Hotel-Restaurant, Retail Outlets for Dairy-Ice Cream, Pharmaceuticals, Logistics, Plastic, Chemical Processing industry and Biological Research industries.. These cutting edge products are manufactured at class apart, ISO certified plant that are aided with well-developed infrastructure and process to ensure highest quality.

Ice Make is committed to help customers meet the demands of complex prescription for food safety and energy efficiency and to ensure that we offer them competitive pricing, professional advice and innovative design.

All products are built for performance, reliability and durability, even in the toughest and most difficult conditions.





The Core of **TRUST SATISFACTION**

Your **Trusted Cooling Partner** for the preserve ahead

"Satisfaction raised through an intellectual comparison of after-sales service and product delivered with quality that shows our successful long-term trusted business relationships with clients"

Why Ice Make?

We are in the business of exploring new frontiers with innovative ideas in providing complete cooling solutions for total customer satisfaction.



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COLD ROOM

Ice Make is a leading manufacturer and supplier of highly efficient Cold Room and Storage that is easy to operate and better in performance. This tailor-made product range is built with premium quality range of material in standard size and customised specification to fulfil our customer needs. These Cold Rooms are well insulated, highly efficient and precisely designed to cater to a wide temperature range from +15°C to -40°C.



Applications: Industries such as


Fruits & Vegetables


Pharmaceutical & Chemical


Ice Cream


Dry Fruits & Spices


Meat & Fish


Bakery Products


Frozen Food


Floriculture

Total Cold Chain Solution



Technical Specification

Thickness : 60, 80,100, 125 & 150 mm

Surface Material : GI Pre-Painted Sheet, S.S. Sheet (Grade 304/316 & Surface finish 2B & Matt), GI Plain Sheet (Inside & Outside different metal can be placed as per your requirement)

Floor : (A) For Kota stone or Concrete floor, Puf Slab with both side tar felt coated.

(B) For Kota Stone or Concrete floor, Puf Panel with Metal Surface at Both Side. Where Panel fits together with camlock to prevent heat loss.

(C) Aluminium Chequered Plate with Marine Ply.

(D) FRP Clading on floor

Door : (A) Flush Type Swing Door with PVC Profile, Imported Hardware, Push Type Gasket & replaceable heater for easy door operation & long life
Opening Size: 24"x66", 30"x72", 34"x78", 46"x84" (Door Opening W x H)

(B) Overlap Type Swing door with frame and metal covering by using heavy duty hardware for better strength
Opening Size: 46"x84", 49"x84", 54"x84", 60"x90" (Door Opening W x H)

Room Size : (A) Length & Width Minimum 1.78 mt and Bigger size 0.146 mt in multiples as per requirement

(B) Standard Height for 60 mm thickness 2.47 mt, 80 mm-2.51 mt, 100 mm-2.55 mt, 125 mm-2.60 mt, 150 mm-2.65 mt. Single Panel Height upto 12 mt and above as per your requirement.

Accessories



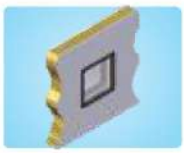
Sliding Door:-
Opening Size
Width : 1000/1200/1500/1800 mm
Height : 2100 to 3000 mm
Thickness : 80 mm
Panel Surface Material : Both side GIPP or S.S as per requirement



Hatch Window:-
Opening Size (WxH): 20"x20", 24"x24", 30"x30"
Different size hatch window with sturdy frame for easy material movement in high turnover ice cream and dairy industries.



Pressure Ventilator:-
Tri-action pressure ventilator with/without heater to balance pressure and vacuum inside cold room.



View Port:-
12"x12" View port with three layer argon filled, toughen glass which can be visible up to 2°C inside the cold room.



Strip Curtain:-
Semi transparent PVC strip curtain for negative temperature application to prevent heat loss and ultimately the power.



Wall Guard



Door Accessories



LED Vapour proof bulk head



High-Low Temperature Alarm



Inside Safety Alarm



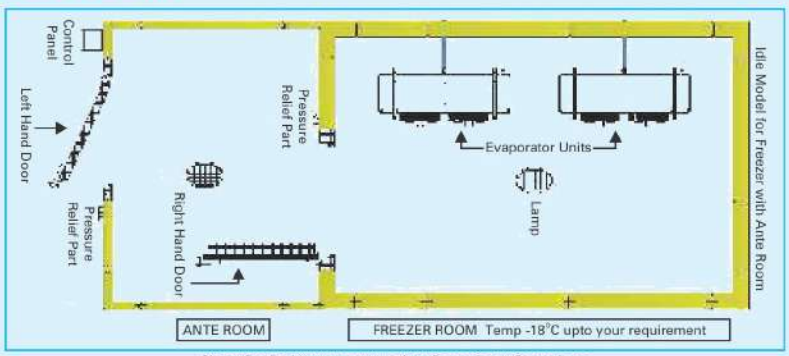
Door with Kick Plate



Ramp with Chequered Plate

3D / Plan View of Ice Cream Cold Room

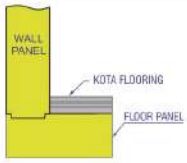


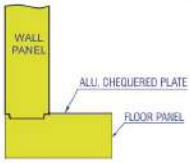


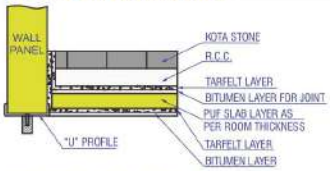
Note : For Positive temperature Ante Room is not Compulsory

Panel Construction Configuration

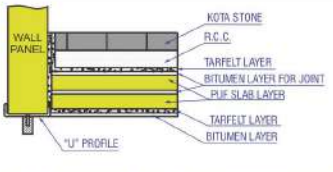
Types of Cold Room Floor







FOR SINGLE PUF SLAB - 60MM, 80MM & 100 MM
(AS PER REQUIRED ROOM FLOOR HEIGHT)



FOR DOUBLE PUF SLAB - 150 MM (60+80) & 125 MM (60+60)
(AS PER REQUIRED ROOM FLOOR HEIGHT)

Vertical & Top Panel Joining Method

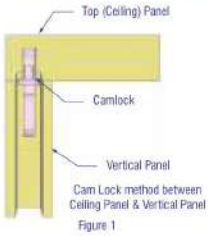


Figure 1

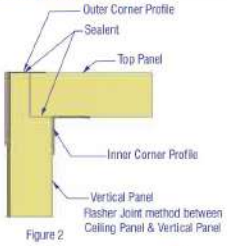


Figure 2

Corner–Vertical Panel Joining Method

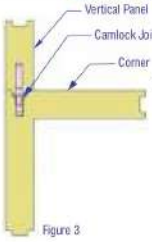


Figure 3

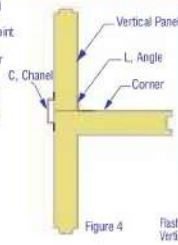


Figure 4

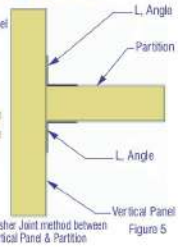
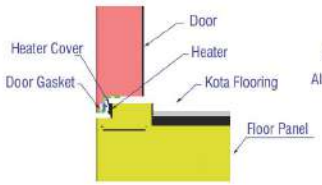
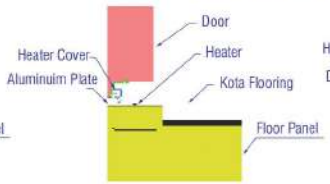
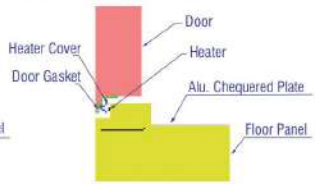


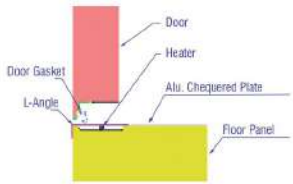
Figure 5

Door on Floor Panel Design

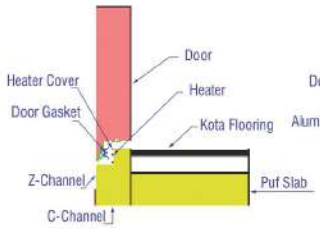


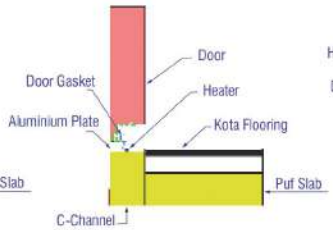


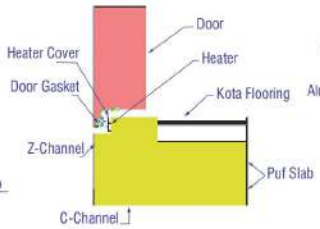


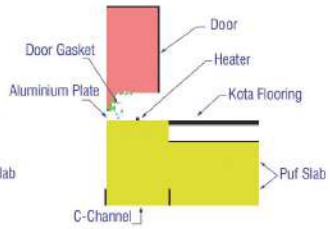


"C" & Z Channel Door Design

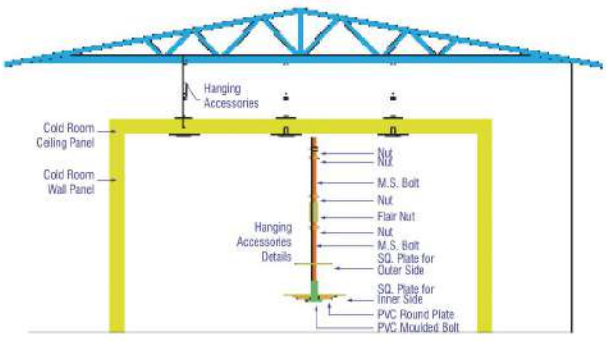




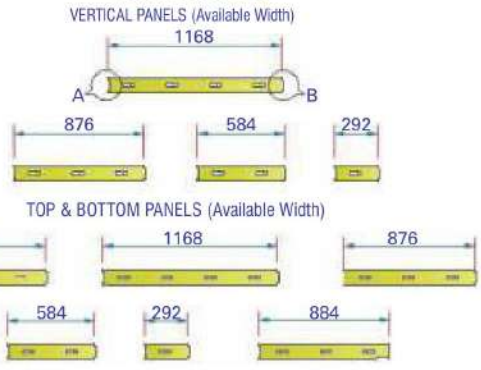




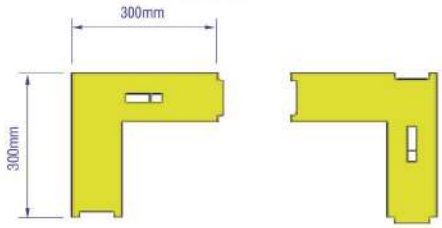
Ceiling Panel Hanging Method



Panel Width Size



Corners



Panel Joints



Available Cold Room Size

Standard Length & Width (with common thickness)

Inch	Feet	Meter	Inch	Feet	Meter	Inch	Feet	Meter	Inch	Feet	Meter	Inch	Feet	Meter
41.25"	3' - 5.25"	1.05	282.75"	23' - 6.75"	7.18	524.25"	43' - 8.25"	13.32	765.75"	63' - 9.75"	19.45	1007.25"	83' - 11.25"	25.58
47"	3' - 11"	1.19	288.5"	24' - 0.5"	7.33	530"	44' - 2"	13.46	771.5"	64' - 3.5"	19.6	1013"	84' - 5"	25.73
52.75"	4' - 4.75"	1.34	294.25"	24' - 6.25"	7.47	535.75"	44' - 7.75"	13.61	777.25"	64' - 9.25"	19.74	1018.75"	84' - 10.75"	25.88
58.5"	4' - 10.5"	1.49	300"	25'	7.62	541.5"	45' - 1.5"	13.75	783"	65' - 3"	19.89	1024.5"	85' - 4.5"	26.02
64.25"	5' - 4.25"	1.63	305.75"	25' - 5.75"	7.77	547.25"	45' - 7.25"	13.9	788.75"	65' - 8.75"	20.03	1030.25"	85' - 10.25"	26.17
70"	5' - 10"	1.78	311.5"	25' - 11.5"	7.91	553"	46' - 1"	14.05	794.5"	66' - 2.5"	20.18	1036"	86' - 4"	26.31
75.75"	6' - 3.75"	1.92	317.25"	26' - 5.25"	8.06	558.75"	46' - 6.75"	14.19	800.25"	66' - 8.25"	20.33	1041.75"	86' - 9.75"	26.46
81.5"	6' - 9.5"	2.07	323"	26' - 11"	8.2	564.5"	47' - 0.5"	14.34	806"	67' - 2"	20.47	1047.5"	87' - 3.5"	26.61
87.25"	7' - 3.25"	2.22	328.75"	27' - 4.75"	8.35	570.25"	47' - 6.25"	14.48	811.75"	67' - 7.75"	20.62	1053.25"	87' - 9.25"	26.75
93"	7' - 9"	2.36	334.5"	27' - 10.5"	8.5	576"	48'	14.63	817.5"	68' - 1.5"	20.76	1059"	88' - 3"	26.9
98.75"	8' - 2.75"	2.51	340.25"	28' - 4.25"	8.64	581.75"	48' - 5.75"	14.78	823.25"	68' - 7.25"	20.91	1064.75"	88' - 8.75"	27.04
104.5"	8' - 8.5"	2.65	346"	28' - 10"	8.79	587.5"	48' - 11.5"	14.92	829"	69' - 1"	21.06	1070.5"	89' - 2.5"	27.19
110.25"	9' - 2.25"	2.8	351.75"	29' - 3.75"	8.93	593.25"	49' - 5.25"	15.07	834.75"	69' - 6.75"	21.2	1076.25"	89' - 8.25"	27.34
116"	9' - 8"	2.95	357.5"	29' - 9.5"	9.08	599"	49' - 11"	15.21	840.5"	70' - 0.5"	21.35	1082"	90' - 2"	27.48
121.75"	10' - 1.75"	3.09	363.25"	30' - 3.25"	9.23	604.75"	50' - 4.75"	15.36	846.25"	70' - 6.25"	21.49	1087.75"	90' - 7.75"	27.63
127.5"	10' - 7.5"	3.24	369"	30' - 9"	9.37	610.5"	50' - 10.5"	15.51	852"	71'	21.64	1093.5"	91' - 1.5"	27.77
133.25"	11' - 1.25"	3.38	374.75"	31' - 2.75"	9.52	616.25"	51' - 4.25"	15.65	857.75"	71' - 5.75"	21.79	1099.25"	91' - 7.25"	27.92
139"	11' - 7"	3.53	380.5"	31' - 8.5"	9.66	622"	51' - 10"	15.8	863.5"	71' - 11.5"	21.93	1105"	92' - 1"	28.07
144.75"	12' - 0.75"	3.68	386.25"	32' - 2.25"	9.81	627.75"	52' - 3.75"	15.94	869.25"	72' - 5.25"	22.08	1110.75"	92' - 6.75"	28.21
150.5"	12' - 6.5"	3.82	392"	32' - 8"	9.96	633.5"	52' - 9.5"	16.09	875"	72' - 11"	22.22	1116.5"	93' - 0.5"	28.36
156.25"	13' - 0.25"	3.97	397.75"	33' - 1.75"	10.1	639.25"	53' - 3.25"	16.24	880.75"	73' - 4.75"	22.37	1122.25"	93' - 6.25"	28.51
162"	13' - 6"	4.11	403.5"	33' - 7.5"	10.25	645"	53' - 9"	16.38	886.5"	73' - 10.5"	22.52	1128"	94'	28.65
167.75"	13' - 11.75"	4.26	409.25"	34' - 1.25"	10.39	650.75"	54' - 2.75"	16.53	892.25"	74' - 4.25"	22.66	1133.75"	94' - 5.75"	28.8
173.5"	14' - 5.5"	4.41	415"	34' - 7"	10.54	656.5"	54' - 8.5"	16.68	898"	74' - 10"	22.81	1139.5"	94' - 11.5"	28.94
179.25"	14' - 11.25"	4.55	420.75"	35' - 0.75"	10.69	662.25"	55' - 2.25"	16.82	903.75"	75' - 3.75"	22.96	1145.25"	95' - 5.25"	29.09
185"	15' - 5"	4.7	426.5"	35' - 6.5"	10.83	668"	55' - 8"	16.97	909.5"	75' - 9.5"	23.1	1151"	95' - 11"	29.24
190.75"	15' - 10.75"	4.85	432.25"	36' - 0.25"	10.98	673.75"	56' - 1.75"	17.11	915.25"	76' - 3.25"	23.25	1156.75"	96' - 4.75"	29.38
196.5"	16' - 4.5"	4.99	438"	36' - 6"	11.13	679.5"	56' - 7.5"	17.26	921"	76' - 9"	23.39	1162.5"	96' - 10.5"	29.53
202.25"	16' - 10.25"	5.14	443.75"	36' - 11.75"	11.27	685.25"	57' - 1.25"	17.41	926.75"	77' - 2.75"	23.54	1168.25"	97' - 4.25"	29.67
208"	17' - 4"	5.28	449"	37' - 5.5"	11.42	691"	57' - 7"	17.55	932.5"	77' - 8.5"	23.69	1174"	97' - 10"	29.82
213.75"	17' - 9.75"	5.43	455.25"	37' - 11.25"	11.56	696.75"	58' - 0.75"	17.7	938.25"	78' - 2.25"	23.83	1179.75"	98' - 3.75"	29.97
219.5"	18' - 3.5"	5.58	461"	38' - 5"	11.71	702.5"	58' - 6.5"	17.84	944"	78' - 8"	23.98	1185.5"	98' - 9.5"	30.11
225.25"	18' - 9.25"	5.72	466.75"	38' - 10.75"	11.86	708.25"	59' - 0.25"	17.99	949.75"	79' - 1.75"	24.12	1191.25"	99' - 3.25"	30.26
231"	19' - 3"	5.87	472.5"	39' - 4.5"	12	714"	59' - 6"	18.14	955.5"	79' - 7.5"	24.27	1197"	99' - 9"	30.4
236.75"	19' - 8.75"	6.01	478.25"	39' - 10.25"	12.15	719.75"	59' - 11.75"	18.28	961.25"	80' - 1.25"	24.42	1202.75"	100'-2.75"	30.54
242.5"	20' - 2.5"	6.16	484"	40' - 4"	12.29	725.5"	60' - 5.5"	18.43	967"	80' - 7"	24.56	1208.50"	100'-8.5"	30.69
248.25"	20' - 8.25"	6.31	489.75"	40' - 9.75"	12.44	731.25"	60' - 11.25"	18.57	972.75"	81' - 0.75"	24.71	1214.25"	101'-2.25	30.84
254"	21' - 2"	6.45	495.5"	41' - 3.5"	12.59	737"	61' - 5"	18.72	978.5"	81' - 6.5"	24.85	1220"	101'-8.0"	30.98
259.75"	21' - 7.75"	6.6	501.25"	41' - 9.25"	12.73	742.75"	61' - 10.75"	18.87	984.25"	82' - 0.25"	25	1225.75"	102'-1.75"	31.13
265.5"	22' - 1.5"	6.74	507"	42' - 3"	12.88	748.5"	62' - 4.5"	19.01	990"	82' - 6"	25.15	1231.5"	102'-7.5"	31.28
271.25"	22' - 7.25"	6.89	512.75"	42' - 8.75"	13.02	754.25"	62' - 10.25"	19.16	995.75"	82' - 11.75"	25.29	1237.25"	103'-1.25"	31.42
277"	23' - 1"	7.04	518.5"	43' - 2.5"	13.17	760"	63' - 4"	19.3	1001.5"	83' - 5.5"	25.44	1243"	103'-7"	31.57

Standard Height (with common thickness)

Height	Thickness	60 mm					
		PUF Slab			Alu. Checkered		
	Wall Panel	Inch	Feet	Meter	Inch	Feet	Meter
St. Height-1	92.5	94.9	7'-10.9"	2.4101	97.22	8'-1.2"	2.47
St. Height-2	112	114.4	9'-6.4"	2.91	116.7	9'-8.7"	2.96
Maximum single panel	472.5	242.2	20'-2.4"	6.1576	244.7	20'-4.7"	6.22

Height	Thickness	80 mm					
		PUF Slab			Alu. Checkered		
	Wall Panel	Inch	Feet	Meter	Inch	Feet	Meter
St. Height-1	92.5	95.6	7'-11.6"	2.43	98.8	8'-2.8"	2.51
St. Height-2	112	115.1	9'-7.1"	2.92	118.3	9'-10.3"	3
Maximum single panel	472.5	243.1	20'-3.1"	6.18	246.3	20'-6.3"	6.26

Height	Thickness	100 mm					
		PUF Slab			Alu. Checkered		
	Wall Panel	Inch	Feet	Meter	Inch	Feet	Meter
St. Height-1	92.5	96.4	8'-0.4"	2.45	100.39	8'-4.4"	2.55
St. Height-2	112	115.9	9'.7.9"	2.94	119.9	9'-11.9"	3.05
Maximum single panel	472.5	243.9	20'-3.9"	6.20	247.9	20'-7.9"	6.30

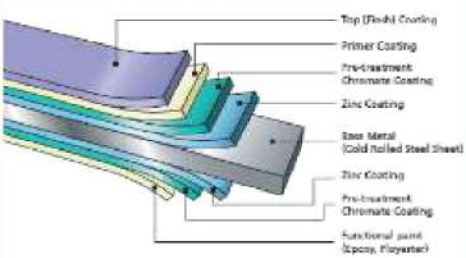
Height	Thickness	125 mm					
		PUF Slab			Alu. Checkered		
	Wall Panel	Inch	Feet	Meter	Inch	Feet	Meter
St. Height-1	92.5	97.4	8'-1.4"	2.48	102.34	8'-6.3"	2.60
St. Height-2	112	116.9	9'.8.9"	2.97	121.8	10'-1.8"	3.09
Maximum single panel	472.5	244.9	20'-4.9"	6.22	249.8	20'-9.8"	6.35

Height	Thickness	150 mm					
		PUF Slab			Alu. Checkered		
	Wall Panel	Inch	Feet	Meter	Inch	Feet	Meter
St. Height-1	92.5	98.4	8'-2.4"	2.50	104.3	8'-8.3"	2.65
St. Height-2	112	117.9	9'-9.9"	2.99	123.8	10'-3.8"	3.14
Maximum single panel	472.5	245.9	20'-5.9"	6.25	251.8	20'-11.8"	6.40

PUF Thickness (mm)	Recomm-ended Temp. Degree @ 32°C Ambient Temp.	Weight						Thermal Conductivity 'K' Value W/mK	Thermal Transmittance 'U' Value W/m²K	Thermal Resistance 'R' Value m²K/W	Thermal Transmittance 'U' Value K.Cal/m²K	Thermal Resistance 'R' Value m²K/K.Cal
		Wall & Ceiling NOTE for wall & ceiling panel: The above is actual panel weight with both side GIPP 0.4 mm thickness and it includes accessories i.e cam locks, spacers, PVC gasket etc..		Floor Panel								
				Floor bare slab		Alu. Checkered Floor NOTE for Alu. Checkered floor: The above is actual panel weight with one side GIPP 0.4 mm thk. & other side 9 mm marine ply with 2 mm Al. checkered plate and it includes accessories i						
		Kg/Sq.ft	Kg/Sq.mtr	Kg/Sq.ft	Kg/Sq.mtr	Kg/Sq.ft	Kg/Sq.mtr					
60	+20 to +2	0.9	9.67	0.25	3	2.5	27	0.022	0.3521	2.84	0.3018	3.31
80	+2 to -8	1	10.82	0.35	4	2.6	28	0.022	0.2679	3.73	0.2296	4.36
100	-8 to -18	1.06	11.3	0.425	5	2.7	29	0.022	0.2163	4.62	0.1854	5.39
125	-18 to -27	1.2	12.72	0.525	6	2.85	31	0.022	0.1721	5.81	0.1475	6.78
150	-27 to -50	1.25	13.45	0.65	7	3	32	0.022	0.1459	6.85	0.1251	7.99

	Thermal Flux (Heat loss per m² wall area) at different △T									
	W/m²					K.cal / m²				
Panel Thickness (mm)	60	80	100	125	150	60	80	100	125	150
Temp. Difference °C										
1	0.3521	0.2679	0.2163	0.1721	0.1459	0.3018	0.2296	0.1854	0.1475	0.1251
10	3.5205	2.6797	2.1631	1.7556	1.4596	3.0174	2.2968	1.8540	1.5047	1.2510
15	5.2808	4.0196	3.2446	2.6335	2.1894	4.5262	3.4452	2.7810	2.2572	1.8766
20	7.0411	5.3595	4.3262	3.5113	2.9192	6.0349	4.5936	3.7080	3.0095	2.5021
25	8.8014	6.6993	5.4077	4.3891	3.6491	7.5437	5.7419	4.6350	3.7619	3.1276
30	10.5617	8.0391	6.4893	5.2670	4.3789	9.0524	6.8903	5.5620	4.5143	3.7531
35	12.3219	9.3790	7.5708	6.1447	5.1087	10.5611	8.0387	6.4890	5.2667	4.3786
40	14.0822	10.7189	8.6524	7.0226	5.8384	12.0699	9.1871	7.4159	6.0191	5.0041
45	15.8424	12.0587	9.7340	7.9004	6.5682	13.5786	10.3356	8.3430	6.7714	5.6296
50	17.6028	13.3986	10.8155	8.7782	7.2980	15.0873	11.4840	9.2699	7.5238	6.2551
55	19.3631	14.7385	11.8971	9.6561	8.0278	16.5961	12.6324	10.1970	8.2762	6.8806
60	21.1239	16.0784	12.9786	10.5339	8.7576	18.1053	13.7808	11.1239	9.0286	7.5062
65	22.8836	17.4182	14.0602	11.4117	9.4874	19.6135	14.9292	12.0510	9.7810	8.1317
70	24.6439	18.4382	15.1417	12.2896	10.2172	21.1223	15.8034	12.9779	10.5334	8.7572
80	28.1644	21.4377	17.3048	14.0452	11.6769	24.1397	18.3743	14.8320	12.0382	10.0082

Technical Specifications of PUF Panels

Sr.No.	PARAMETERS	DESCRIPTION
1	Average PUF Density	40 ± 2 kg/m^3
2	PUF Blowing Agent	Cyclopentane (CFC & HCFC Free)
3	Insulation Material	Polyurethane foam
4	Temperature range	+90°C to -60°C
5	Panel Type	Discontinuous type with camlock
6	M.O.C. of Camlocks	Body material of Camlock is HIPS & male latch is made of GI sheet. Camlock joins panels in easy assembly and panel-to-panel joints are leakage resistant
7	Type of Panel joints	Tongue & Groove Joint with Cam lock and all side rubber gaskets
8	Corner Panel Available	12" x 12" x 162" (maximum Length)
9	Length of Wall & ceiling Panel	2032 mm to 12000 mm 80" to 472.5" inch
10	Ceiling Suspension items with specs	MS bolt with thermal breaking cap. PVC Round Plate 6 mm thickness x 90 mm diameter Aluminium/MS washer plate for load distribution Anchor bolt (MS) hanging bolt with 40 mm washer 2.0 mm thickness Sealing to panel MS road Clamp with an insulated crown.
11	Closed cell content	90 to 95%
12	Vapour Permeability	5.5mg/PASM
13	Water Absorption	Less than 2%
14	Fire Grade Polyurethane PUF Panel	FR Grade B3 (Fire Retardant Grade)
15	Compressive Strength at 10% Deformation	* 175 kN/mm2 (For PUF only)
16	Tensile Strength	0.11 N/mm2 (For PUF only)
17	Adhesive Strength (Foam to Sheet)	0.45 kg/cm2
18	Dimension Stability	Less than 2%
19	Panel Facing Availability	With Rib or plain on Demand (Ribs are provided for more strength)
20	Wall & Ceiling Panel facing Material	• Pre Painted Galvanised sheet , Thickness 0.4mm • Plain Galvanised sheet , Thickness 0.4mm • S.S. Sheet, Grade 316, 0.50mm, finish 2B/0.6 mm matt No.4 • S.S. Sheet, Grade 304, 0.50mm, finish 2B/0.6 mm matt No.4
21	Floor Panel Facing Material	• PUF Panel with both side tarfelt sheet. • PUF Panel with both side 0.4mm GIPP sheet. • PUF Panel with outside GIPP sheet, inside 9 mm thick marine ply with 2 mm thick Aluminium/SS Chequered Plate
22	Specification for Colour Coating 	**A.** RAL No. : Ice Make White **B.** Indian Standard code Colour Coating : IS 14246 Galvanizing : IS 277 Base Metal : IS 513 **C.** Organic Coating : Type RMP (Polyester) **D.** Zinc Coating : 120 GSM **E.** Top Primer (Thickness in micron 'μ') : 5 +/- 1 μ **F.** Top Coat (Thickness in micron 'μ') : 20 +/- 1 μ **G.** Back Primer (Thickness in micron 'μ') : 4 +/- 1 μ **H.** Back Coat (Thickness in micron 'μ') : 4 +/- 1 μ **I.** Guard Film (Thickness in micron 'μ') : 40 +/- 3 μ **J.** Salt Spray Test /Humidity Test : 750 hrs. /1000 hrs.

SLIDING DOOR

ICE MAKE Sliding Doors are delivered as complete units, easy to mount with all fittings, sliding rail and bottom guide roller etc. Sliding Doors are designed for tough commercial environments such as Cold Stores, Incubation Chamber and Blast Chiller.

ICE MAKE Sliding Doors are very easy to operate, greater in efficiency and better in performance; also in terms of safety, long service life and smooth in operation everyday with one hand.
ICE MAKE Sliding Door has sealed upper surface to resist dirt build up, very easy to clean and very easy to close and reliably opens.



Handle with Clamp



L Clamp

Wheel



Top Stopper

Side Stopper



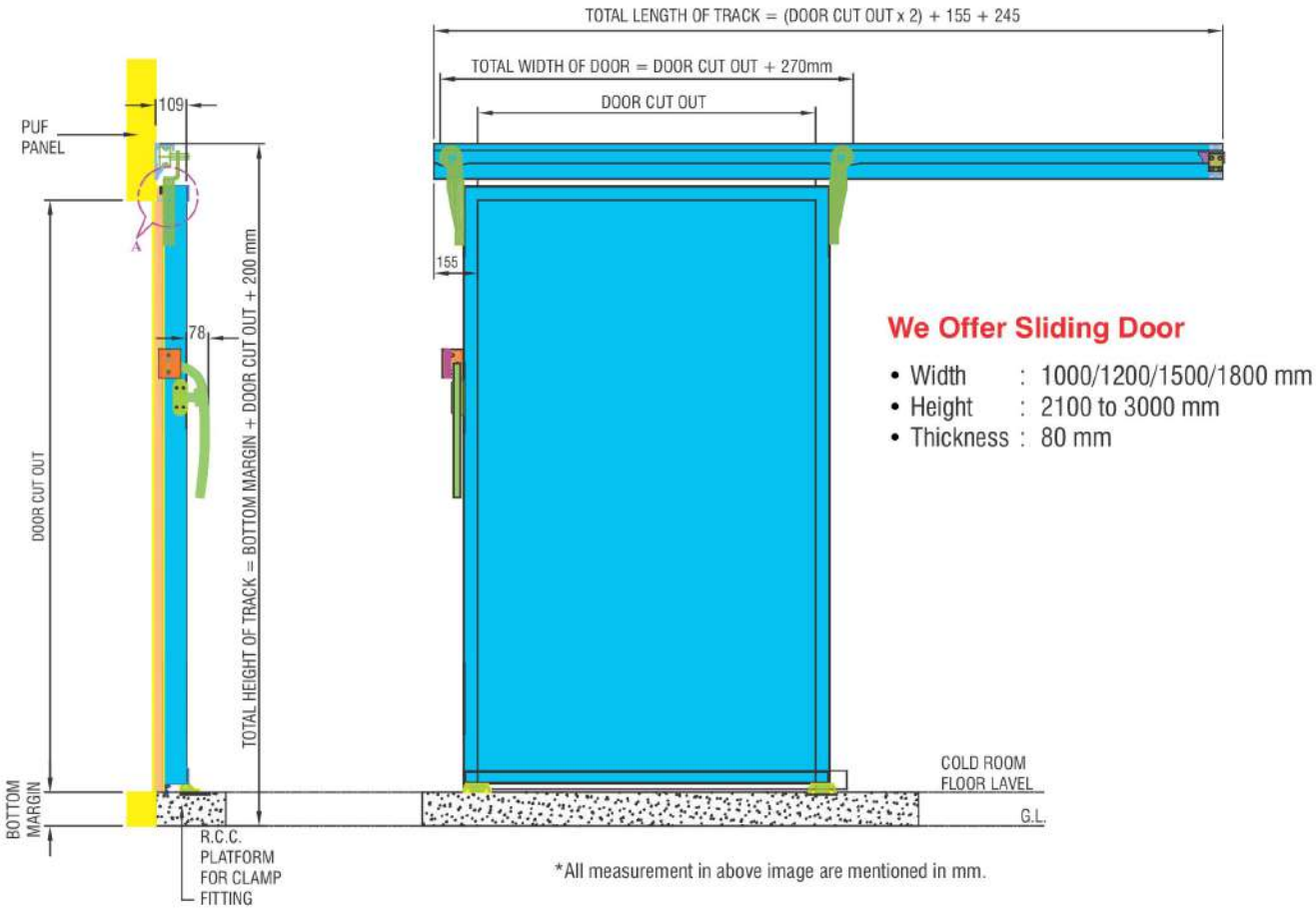
Bottom Support Clamp

Ice Make Sliding Door Features:

- Heavy duty imported hardware
- Formed in place door panels
- S.S door frame
- Heavy duty track and wheel for durability
- Door guide to keep door on track
- Gasket to seal off air leak
- Durable and long service life, corrosion and Wear-resistant
- Easy maintenance & robust in design
- Outstanding ease of operation by manual
- High standard of hygiene and easy-to-clean surfaces

Technical Specifications

Door Panel	80 mm thick (rigid Polyurethane foam)
Door Frames	Assembled in Panel
Surface Finish	The surface finish is available in a range of RAL No. Ice Make White or its nearest colours, with Pre-painted sheet, S.S sheet (304/316), with GI Pre painted sheet 0.4 /0.7 mm, SS sheet 304 (with 2B finish, 0.5 mm & with matt finish 0.6 mm).
Insulation	Injected Polyurethane insulation with, • Thermal Conductivity 0.022 W/m^{2}K • Density 40 ± 2 kg/m^3 • Tensile Strength 0.11 N/mm^2 • Vapour permeability 5.5 mg/PASM
Hardware	• Anodized Aluminium Upper Guide rail • Availability of floor (F) guide Clamp • Brackets in reinforced composite material and zinc-coated metal • Outside opening devices with lever handle • Mounting on pre-fabricated panel • Simple and quick cleaning surface • Indoor Leaf with Replaceable Single Gasket



We Offer Sliding Door

- Width : 1000/1200/1500/1800 mm
- Height : 2100 to 3000 mm
- Thickness : 80 mm

*All measurement in above image are mentioned in mm.

SOLAR COLD ROOM

ICE MAKE

Backup up to 30 hours

Low Maintenance

Solar Powered Stand-alone

Grid & DG Set Hybrid

No Battery for Refrigeration

No Running Cost on Solar Supply

Portable / Easy to Shift



Ice Make has gained continuous appreciation as the prime organization offering Commercial & Industrial Refrigeration Equipment catering to Industries like Dairy, Ice Cream, Food Processing, Fruit Ripening, Horticulture / Floriculture, Hotels-Restaurants, Pharmaceuticals, Logistics, Plastic and Chemical Industry etc. This appreciation encouraged us to develop the innovative Cold Storage powered by solar stand-alone for improving storage quality and reducing wastage of horticulture and floriculture produce.

Most of the Horticultural/Floricultural produce requires a cooling temperature between 0°C to 15°C for safe storage and transient purposes. In the absence of cold storage and related cold chain facilities, the farmers are forced to sell their produce immediately after harvest which results in overabundance and low price realization.

Stand-alone Solar Power is one of the best solutions for operating small cold storage system in rural areas where there is certain limit of power load. Solar energy based refrigeration system is quite relevant to Indian weather because it is blessed with a good amount of solar energy in most parts of the country, throughout the year. This rich source of endless solar energy is the main power source for the Solar Cold Room built by Ice Make and it gives a 24 Hours backup with no door opening.

Ice Make Solar Cold Room is a Hybrid Cold Room designed to use throughout the year, even when there is no sunlight. This Solar Cold Room can be used with the alternate power source during the absence of Sunlight i.e. Electric Power and/or DG Power.

The Plug & Play feature of this Solar Cold Room makes it portable. Being portable, this Cold Room can be shifted anytime anywhere as per your convenience with no installation cost. Additionally the maintenance is very low and there is no running cost while operating through solar stand-alone. These characteristics make this Solar Cold Room best suited & affordable for farmers.

We offer Solar based Cold Storage

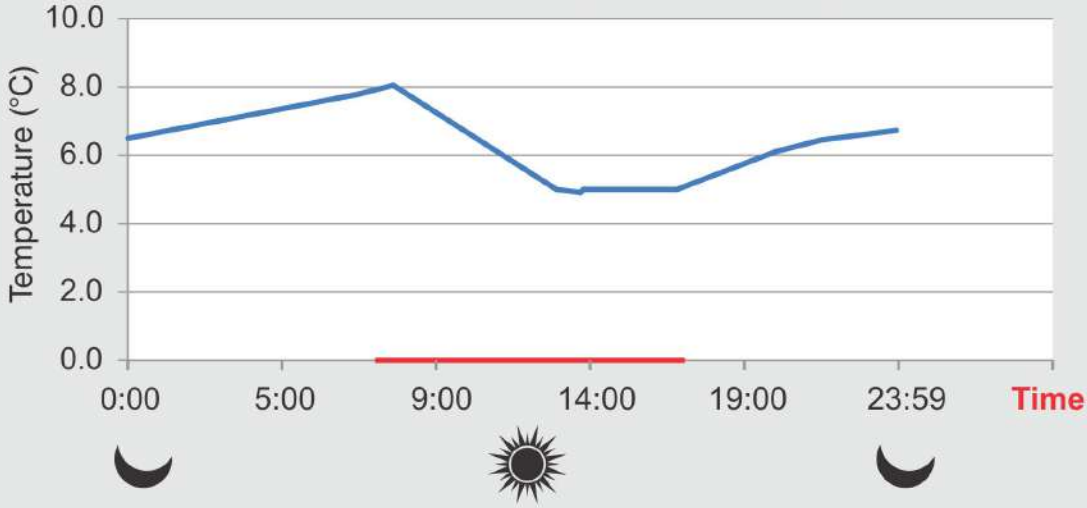
- Product capacity : 2MT and above
- Temperature range of 4 °C and above*
- Thermal energy Battery backup up to 30 hours
- IOT enabled Predictive Maintenance
- Grid/DG Hybrid: Can run on grid/dg in non solar days
- Even can run during the hours of low sunlight using Inverter Tech

SOLUTIONS TO GO WITH CUSTOMER'S REQUIREMENT

Multiple ways to own

- Upfront purchase
- Lease & Rental
- Community Model

24 Hours Temperature Graph





Fruit | Flower | Vegetable | Dairy | Pharma

GLASS DOOR DISPLAY CHILLER

ICE MAKE Glass Door Display Chiller are suitable for super market and retails shops to preserve and display Dairy items, Cake-pastry, Chocolates, Beverages, Fruits & Vegetables, Flower, Medicines etc. These glass door display chiller are designed and developed with utmost eminence in accordance with the defined parameters of the industry and emerging market demands this classic design combined with up-to-date technology enable timeless and functional display solutions that enable high-quality product presentation. These display chillers are divided into two sections to keep a variety of items as display and storage inside.

Temperature : 2°C to 8°C





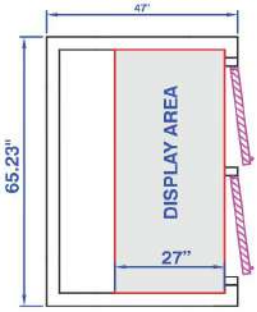
Special Features

- Large Storage with Better Display Area
- Customized size
- Temperature retaining
- Easy maintenance
- Energy efficient
- Easy assemble & dismantle

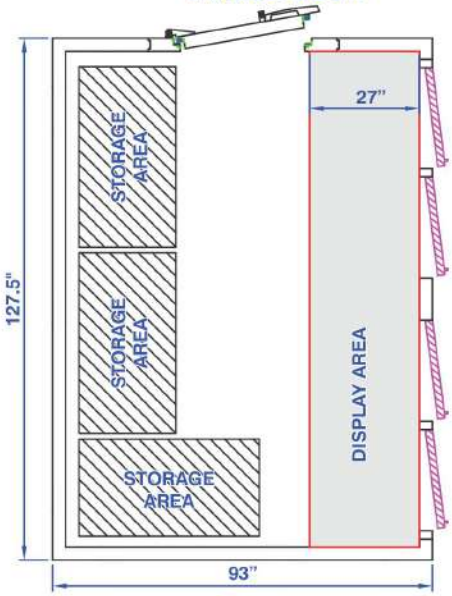
Available Glass Door Display Chiller Size:

Display Area				Display with Storage Area			
No. of Door	Chiller Size (L x W x H)	Volume CFT	Connected Power	No. of Door	Chiller Size (L x W x H)	Volume CFT	Connected Power
2	65.23" x 47" x 98.8"	126	1 / 3 Ph, 1.58 kW	2	65.23" x 93" x 98.8"	269	1 / 3 Ph, 1.68 kW
3	98.75" x 47" x 98.8"	201	1 / 3 Ph, 1.68 kW	3	93.22" x 93" x 98.8"	429	1 / 3 Ph, 1.68 kW
4	127.5" x 47" x 98.8"	264	1 / 3 Ph, 1.68 kW	4	127.5" x 93" x 98.8"	563	1 / 3 Ph, 2.17 kW
5	156.25" x 58.5" x 98.8"	419	1 / 3 Ph, 2.17 kW	5	156.25" x 93" x 98.8"	696	3 Ph, 3.26 kW
6	185" x 58.5" x 98.8"	499	1 / 3 Ph, 2.17 kW	6	185" x 93" x 98.8"	829	3 Ph, 3.26 kW

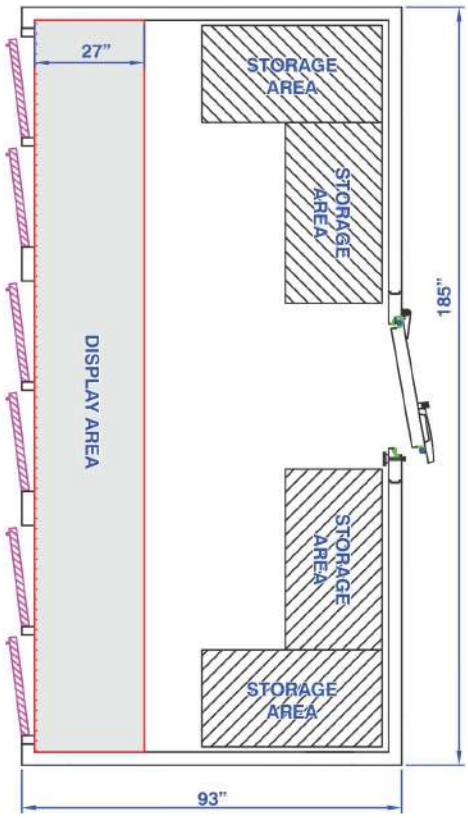
2-GLASS DOOR



4-GLASS DOOR



6-GLASS DOOR



Technical Specification

- **Glass Door Frame Size** : 1401 mm x 1840 mm, **Single Door Opening Size :** 600 mm X 1720 mm.
- Customized glass doors (hinged type) with magnetic door gasket for air tight seal.
- Minimum 2 Glass Door.
- Floor Insulation: Aluminium Chequered plate.
- Digital temperature controller with digital display.
- Insulated perimeter frames designed with forward facing for easy service and maintenance from front of the door.
- Energy efficient frame that reduces heat loss throw it by 30%with added insulation.
- Double layer vacuumed tempered glass to give clear inside view and product protection.
- For specific arrangement of products.
- Adjustable Stainless Steel Shelves System for specific arrangements of product.
- Provision for special demands such as glass doors, special flooring, silent condensing unit, Shelves system, etc.
- Long durable and perfect construction with balanced refrigeration system.





Condensing Unit

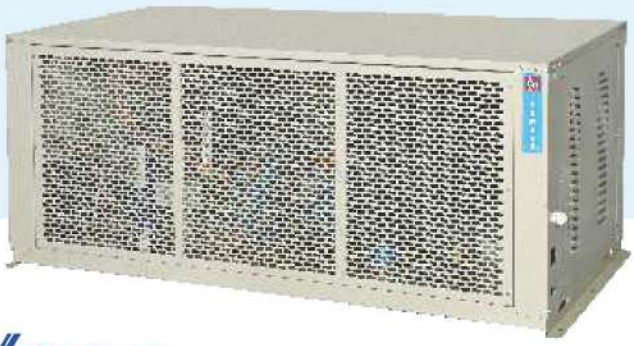
Air-cooled & Water-cooled



Slim Type









Screw Type Condensing Unit

Features

- High cooling with low power consumption
- Designed for high ambient condition up to 55°C
- Condensing coil with inner grooved copper tubes & aluminum fins
- Efficient fans with external motors for single phase and three phase
- High & low pressure cut-out including mounting brackets, wired to terminal strip
- Large size filter drier, moisture indicator, solenoid valve, oil separator and accumulator
- Shell & tube type condenser with high cooling capacity for high ambient conditions

Rack System - An Advanced Refrigeration Technology :

- It has advanced power saving system compared to normal refrigeration system
- Rack system will work as per product load variation



Water-Cooled

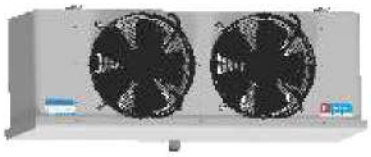
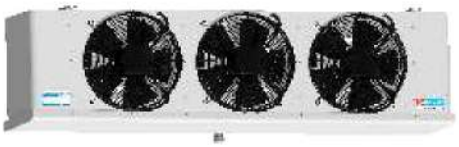


Water-Cooled



Air-Cooled

Evaporator Unit


FEATURES

- **Fan** - High reliability, low-temperature resistance and low noise external rotor fans.
- **Coil** - High efficiency heat exchange with in-line tube system for minimum loss of air flow between fans and large surface area for better cooling.
- **Defrost** - Electrically heating stainless steel pipe is used for long life and it also leak proof.
- **Unit Body** - Aluminium, PU type Powder Coated, Corrosion resistant and nice appearance body.
- **Maintenance** - Compact, adjustable & easily openable side panels for easy installation.

Control Panel



Features

- Temperature Controls
- Phase Preventing
- HP-LP Protection
- Over current protection
- Delay timer



Features

- Phase Preventing
- HP-LP Protection
- Auto Defrost
- Delay Timer
- Over/Under current protection
- High-Low temperature alarm
- Quick freezing option
- Door alarm
- All type of trip alarm & Record Data
- Available in Air & Water cooled system

Dehumidifier

Our portable dehumidifier provide low humidity conditions necessary to maintain a dry environment. It's a simple solution to an old problem. Our Dehumidifier works by forcing condensation on cold surfaces. Humid air is drawn over cooled fins that makes moisture condense on the large cool surface. The condensate is allowed to drain from the fins into a collection tank or is piped to a suitable drainage. Cooled dry air passes through condenser coil and again heated to desired temperature to maintain low relative humidity. large cool surface.



Application:

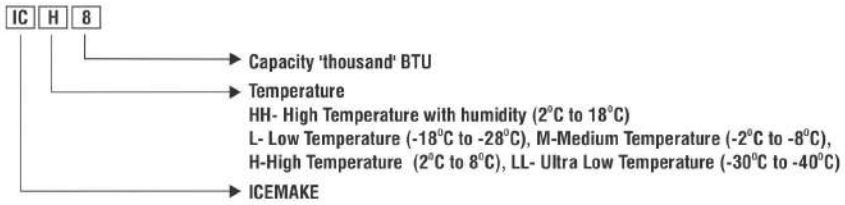
- Sweet storage
- Dehydrated product storage
- Seeds storage
- Bakery product storage and many more

Features

- Auto cut-off according to set RH value
- Automatic defrost
- Top air flow
- Continuous Drainage Option or water collecting tank

particular	Model – 1	Model - 2	Model - 3	Model - 4
Moisture condense capacity (@ 20° C & 60% RH) (L/day)	16	37	60	112
refrigerant	R-134a	R-22	R-22	R-22
Air flow (m3/h)	130	300	600	1000
Connected power (kW)	0.6	1.25	1.5	3
Max power (kW)	0.9	1.6	1.8	3.5
Power supply	1 ph, 220V	1 ph, 220V	1 ph, 220V	1 ph, 220V

Model Code Logic



REFRIGERATION TECHNICAL DATA

Sr. No	Model	Cold Room Volume		Suggested Cooling Capacity					Connected Load kW	Power Supply	Condensor		Refrigerant		Compressor Type		
		CFT	CMT	Watts	H.P.	Cal.	BTU	TR			Air Cooled	Water Cooled	R-404A	R-22	Reciprocating	Scroll	Semi Sealed
FOR HIGH TEMPERATURE @ 2°C TO 8°C																	
1	ICH-8	300	8	2198	2.95	1890	7500	0.62	1.8	1 PH	Yes	No	No	Yes	Yes	No	No
2	ICH-12	500	14	3443	4.62	2961	11748	0.98	2.6/1.93	1/3 PH	Yes	Optional	Yes	Yes	Yes	No	Yes
3	ICH-17	650	18	4899	6.57	4213	16715	1.39	3.1/2.63	1/3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
4	ICH-20	900	25	5634	7.55	4845	19223	1.60	3.08	3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
5	ICH-23	1100	31	7630	10.23	6562	26034	2.17	4.03	3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
6	ICH-40	1900	54	12146	16.28	10446	41442	3.45	6.11	3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
7	ICH-53	2400	68	15776	21.15	13567	53828	4.48	7.9	3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
8	ICH-71	3200	91	20357	27.29	17507	69458	5.78	10.14	3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
9	ICH-85	4000	113	25420	34.08	21861	86733	7.22	12.51	3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
10	ICH-100	5200	147	29886	40.06	25702	101971	8.49	14.89	3 PH	Yes	Optional	Yes	Yes	No	Yes	Yes
11	ICH-145	7000	198	42391	56.82	36456	144638	12.04	21.91	3 PH	Yes	Optional	Yes	Yes	No	Yes	Yes
12	ICH-175	9000	255	51390	68.89	44195	175343	14.60	25.61	3 PH	Yes	Optional	Yes	Yes	No	Yes	Yes
13	ICH-210	11000	311	62586	83.90	53824	213543	17.78	32.86	3 PH	Yes	Optional	Yes	Yes	No	Yes	Yes
FOR MEDIUM TEMPERATURE @ -2°C TO -8°C																	
14	ICM-4	250	7	1250	1.68	1075	4265	0.36	1.31	1 PH	Yes	No	No	Yes	Yes	No	No
15	ICM-9	450	13	2751	3.69	2366	9386	0.78	2.4/1.8	1/3 PH	Yes	Optional	Yes	Yes	Yes	No	Yes
16	ICM-12	600	17	3813	5.11	3279	13010	1.08	2.8/2.51	1/3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
17	ICM-15	800	23	4620	6.19	3973	15763	1.31	3	3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
18	ICM-17	1000	28	4778	6.40	4109	16303	1.36	3.92	3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
19	ICM-28	1700	48	7875	10.56	6773	26870	2.24	4.88	3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
20	ICM-40	2200	62	11722	15.71	10081	39995	3.33	7.48	3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
21	ICM-53	2900	82	15739	21.10	13536	53701	4.47	9.79	3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
22	ICM-65	3750	106	20153	27.01	17332	68762	5.73	11.83	3 PH	Yes	Optional	Yes	Yes	Yes	Yes	Yes
23	ICM-80	4700	133	23950	32.10	20597	81717	6.80	14.24	3 PH	Yes	Optional	Yes	Yes	No	Yes	Yes
24	ICM-112	6400	181	33530	44.95	28836	114404	9.53	21.16	3 PH	Yes	Optional	Yes	Yes	No	Yes	Yes
25	ICM-135	8100	229	39750	53.28	34185	135627	11.29	24.55	3 PH	Yes	Optional	Yes	Yes	No	Yes	Yes
26	ICM-210	9000	255	46600	62.47	40076	158999	13.24	31.68	3 PH	Yes	Optional	Yes	Yes	No	Yes	Yes

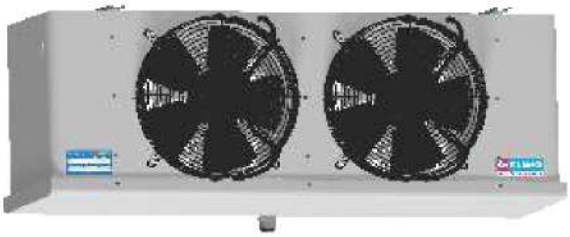
Sr. No	Model	Cold Room Volume		Suggested Cooling Capacity					Connected Load kW	Power Supply	Condensor		Refrigerant		Compressor Type		
		CFT	CMT	Watts	H.P.	Cal.	BTU	TR			Air Cooled	Water Cooled	R-404	R-22	Reci procating	Scroll	Semi Sealed
FOR LOW TEMPERATURE @ -18°C TO -28°C																	
27	ICL-3	300	8	550	0.74	473	1877	0.16	2.5	1 PH	Yes	No	Yes	No	Yes	No	No
28	ICL-4	350	10	1145	1.53	985	3907	0.33	1.98	3 PH	Yes	No	Yes	No	Yes	Yes	No
29	ICL-6	550	16	1788	2.40	1538	6101	0.51	3	3 PH	Yes	Yes	Yes	No	Yes	Yes	Yes
30	ICL-9	850	24	2743	3.68	2359	9359	0.78	4.1	3 PH	Yes	Yes	Yes	No	Yes	Yes	Yes
31	ICL-13	1200	34	3328	4.46	2862	11355	0.95	5.4	3 PH	Yes	Yes	Yes	No	Yes	Yes	Yes
32	ICL-18	1700	48	4743	6.36	4079	16183	1.35	6.9	3 PH	Yes	Yes	Yes	No	Yes	Yes	Yes
33	ICL-26	2500	71	7960	10.67	6846	27160	2.26	8.6	3 PH	Yes	Yes	Yes	No	Yes	Yes	Yes
34	ICL-38	3400	96	11355	15.22	9765	38743	3.23	10.6	3 PH	Yes	Yes	Yes	No	No	Yes	Yes
35	ICL-42	3800	107	10870	14.57	9348	37088	3.09	11.5	3 PH	Yes	Yes	Yes	No	No	Yes	Yes
36	ICL-45	4000	113	12745	17.08	10961	43486	3.62	13.5	3 PH	Yes	Yes	Yes	No	No	No	Yes
37	ICL-50	5000	141	15390	20.63	13235	52511	4.37	15.18	3 PH	Yes	Yes	Yes	No	No	No	Yes
38	ICL-65	6500	184	18690	25.05	16073	63770	5.31	20.29	3 PH	Yes	Yes	Yes	No	No	No	Yes
30	ICL-80	8000	226	21875	29.32	18813	74638	6.21	23.65	3 PH	Yes	Yes	Yes	No	No	No	Yes
40	ICL-105	10500	297	27810	37.28	23917	94888	7.90	32.4	3 PH	Yes	Yes	Yes	No	No	No	Yes
FOR HIGH TEMPERATURE @ 2°C TO 18°C WITH HUMIDITY FOR RIPENING / PRE - COOLING																	
41	ICHH-12	1200	34	5107	6.85	4392	17425	1.45	2.03	1/3 PH	Yes	No	Yes	Yes	Yes	Yes	Yes
42	ICHH-17	1500	42	5033	6.75	4328	17173	1.43	2.53	3 PH	Yes	Yes	Yes	Yes	Yes	Yes	Yes
43	ICHH-20	1700	48	8125	10.89	6988	27723	2.31	3.27	3 PH	Yes	Yes	Yes	Yes	Yes	Yes	Yes
44	ICHH-23	2200	62	10930	14.65	9400	37293	3.11	4.02	3 PH	Yes	Yes	Yes	Yes	Yes	Yes	Yes
45	ICHH-40	3500	99	17648	23.66	15177	60215	5.01	6.15	3 PH	Yes	Yes	Yes	Yes	Yes	Yes	Yes
46	ICHH-53	4500	127	22869	30.66	19667	78029	6.50	8.13	3 PH	Yes	Yes	Yes	Yes	Yes	Yes	Yes
47	ICHH-71	7000	198	29769	39.90	25601	101572	8.46	10.45	3 PH	Yes	Yes	Yes	Yes	Yes	Yes	Yes
48	ICHH-85	8000	226	38056	51.01	32728	129847	10.81	12.97	3 PH	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FOR ULTRA LOW TEMPERATURE -30°C TO -40°C																	
49	ICLL-17	as per heat load		5100	6.84	4386	17401	1.45	5.82	3 PH	Yes	Yes	Yes	No	semi sealed two stage		Yes
50	ICLL-24			7673	10.29	6599	26180	2.18	7.63	3 PH	Yes	Yes	Yes	No			Yes
51	ICLL-39			12025	16.12	10342	41029	3.42	12.26	3 PH	Yes	Yes	Yes	No			Yes
52	ICLL-55			16185	21.70	13919	55223	4.60	15.57	3 PH	Yes	Yes	Yes	No			Yes
53	ICLL-64			19215	25.76	16525	65562	5.46	18.43	3 PH	Yes	Yes	Yes	No			Yes
54	ICLL-72			22315	29.91	19191	76139	6.34	21.05	3 PH	Yes	Yes	Yes	No			Yes
55	ICLL-86			27135	36.37	23336	92585	7.71	25.5	3 PH	Yes	Yes	Yes	No			Yes

* Connected load & Cooling Capacity will be different compressor

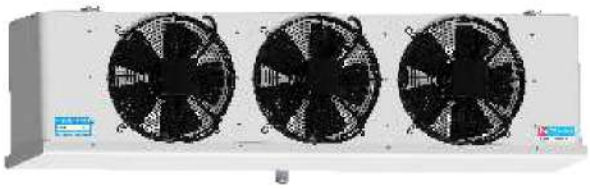
* Optional refrigerants : R134a & R407c also available.



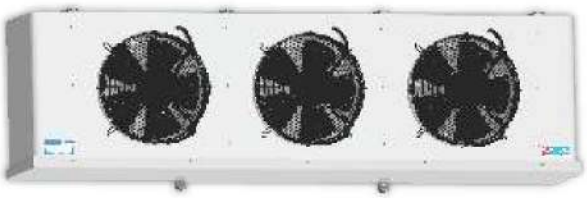
A4007-301H, A7012-301A



A4012-302H, A7021-302A



A4018-303H, A7032-303A



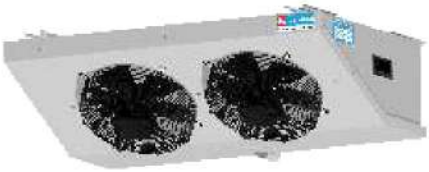
A4024-353H, A4034-353H
A7043-353A, A7060-353A



A4039-452H, A7062-452A



A4058-453H, A7090-453A



CLS709-252A

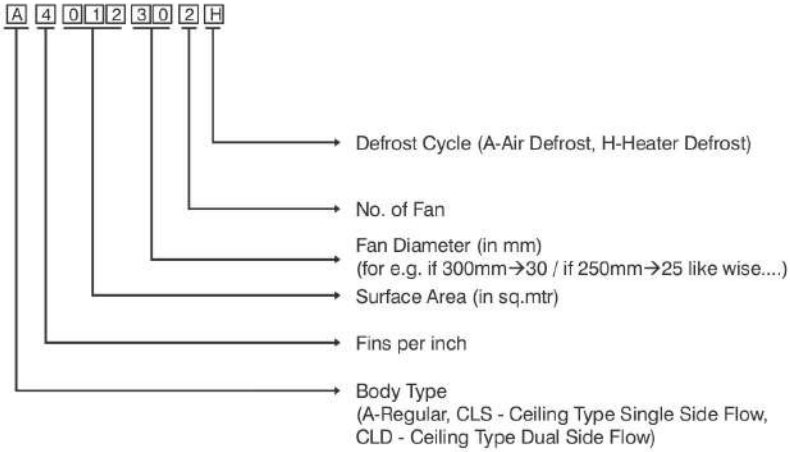


CLD713-252A, CLD719-252A

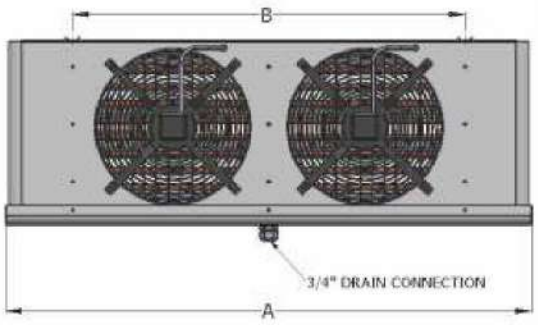


CLD724-253A, CLD729-253A

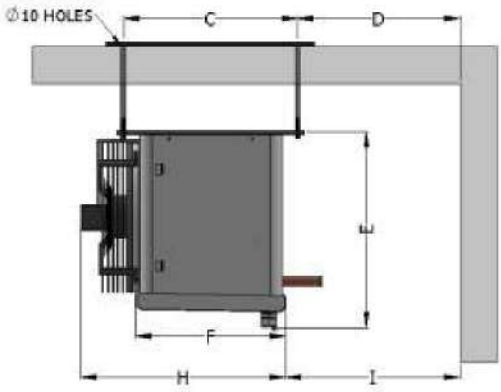
Model Code Logic

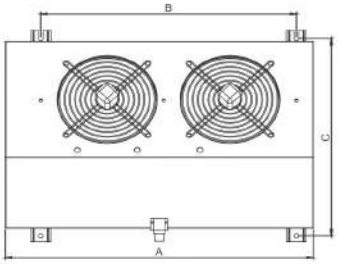


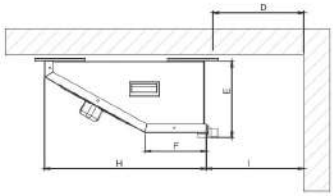
Physical Dimensions



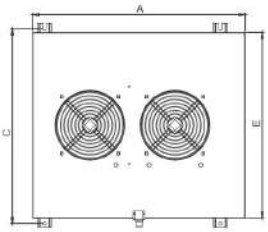
Body Type A

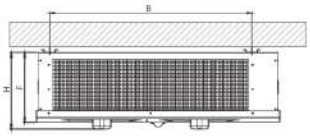






Body Type CLS
(Only SS Canopy)



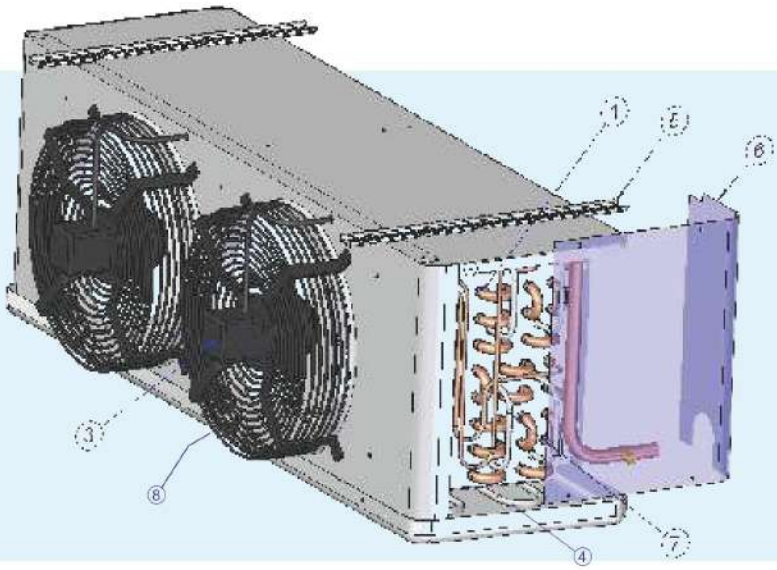


Body Type CLD
(Only SS Canopy)

Physical Perameters	Dimensions (mm) for Model										
	A4007-301H	A4012-302H	A6014-301H	A4018-303H	A4024-353H	A4034-353H	A4039-452H	A4058-453H	CLS709-252A	CLD713-252A	CLD724-253A
	A7012-301A	A7021-302A		A7032-303A	A7043-353A	A7060-353A	A7062-452A	A7090-453A		CLD719-252A	CLD729-253A
A	765	1095	820	1490	1575	1880	1880	2660	825	820	1150
B	485	815	540	1210	1300	1605	1595	2380	685	680	1010
B1	NA	NA	NA	NA	430	535	800	800	NA	NA	NA
C	375	375	375	375	375	375	375	375	530	750	750
D	350	350	350	350	350	350	400	400	300	NA	NA
E	425	425	495	425	495	565	625	425	220	720	720
F	325	325	325	325	325	325	325	325	190	220	220
H	435	435	435	435	435	435	460	460	500	315	315
I	375	375	375	375	375	375	425	425	375	0	0
Unit Weight (kg)	17	26	23	35	46	57	75	115	15	24	36

Features

1) High efficiency heat exchanger with in-line tube system for minimum loss of air flow between fins and with large surface area for long cooling time.
2) Fan Motor and Heater wires are connected in terminal box mounted inside the unit.
3) Fan motors are of leading brand. Insulation Class F, thermally protected against overload and can be used on 230V/50 Hz supplies.
4) Provision given for installing tray heater later.
5) Stainless steel mounting rail.
6) Compact, adjustable & easily openable side panels for easy installation.
7) Heater rods are arranged for better heat distribution in coil. Electric heater rods are inserted into special aluminium sleeve tubes to avoid steam formation.
8) Drainage Connection



Unit Body :

- 1.2mm thick Aluminium, pu type white powder coated, corrosion resistant and nice appearing body.
- Round corner Drain tray and also provided intermediate sheet to avoid condensation.
- Aluminium Die cast, threaded, 3/4" diameter drainage connection.

Finned coil block:

- Aluminium fin, thickness 0.30 mm
- Schrader Valve provided for gas charging.
- Coils are degreased, cleaned and tested with 21 bar Nitrogen pressure and leak tested under water according to standard.

Fan Motor:

- Axial fans with external rotor motor, single phase - 230 V, 50/60 Hz with internally wired thermal contact and wired with internal terminal box.
- Electrical design according to standard.
- Protection class: IP-54, Insulation Class F
- Application range : -35°C to +40°C

Defrost Heating:

- Electric heater rods 230 V sleeve tube with 7-8 mm diameter

Technical Specifications

EVAPORATOR UNIT (Evap. Temp. -8°C to -35°C)									
Sr.	PARAMETERS/MODEL	A4007-301H	A6014-301H	A4012-302H	A4018-303H	A4024-353H	A4034-353H	A4039-452H	A4058-453H
1	Capacity (kW) (at -25°C, DT1=7K)	1.25	2.4	2.5	3.8	5.0	6.8	8.27	11.7
2	Heat Transfer Area (Sq. Mtr)	6.9	14.11	12.15	18.23	24.3	34	39.12	58.12
3	Tube Volume (In Ltr)	2.5	2.9	3.4	5	7	10.21	10.5	15.5
4	Fin Spacing-mm/FPI(Fin Per Inch)	6/4	4/6	6/4	6/4	6/4	6/4	6/4	6/4
5	Fan Diameter & No. of Fans / Ph	300/1/1ph	300/1/Ph	300/2/1ph	300/3/1ph	350/3/1ph	350/3/1ph	450/2/3ph	450/3/3ph
6	Air Flow (m^3/h) / Throw(Mtr)	1560/9	1560/19	3120/9	4680/10	8200/12	8200/12	8600/14	12900/14
7	Motor - Input Value (watts)	87 x 1	87 x 1	87 x 2	87 x 3	150 x 3	150 x 3	320 x 2	320 x 3
8	Motor - Current (Amp.)	0.42 x 1	0.42 x 1	0.42 x 2	0.42 x 3	0.65 x 3	0.65 x 3	0.74 x 2	0.74 x 3
9	Coil & Tray Heater Total (watts)	2500	2000	3750	4250	5000	5000	6550	9450
10	Pipe Inlet & Outlet	1/2 & 3/4	1/2 & 3/4	1/2 & 3/4	1/2 & 7/8	5/8 & 1.1/8	5/8 & 1.1/8	5/8 & 1.1/8	5/8 &1.3/8

EVAPORATOR UNIT (Evap. Temp. 10°C to -8°C)								
Sr. NO	Parameters/Model	A7012-301A	A7021-302A	A7032-303A	A7043-353A	A7060-353A	A7062-452A	A7090-453A
1	Capacity (kW) (at -8°C, DT1= 8 K)	1.75	3.5	5.25	7	9.5	13.5	19
2	Heat Transfer Area (Sq. Mtr)	12.3	21.3	32.1	42.6	59.5	62.45	90.75
3	Tube Volume (In Ltr)	2.5	3.4	5	7	10.21	10.5	15.5
4	Fin Spacing-mm/FPI(Fin Per Inch)	3.6/7	3.6/7	3.6/7	3.6/7	3.6/7	3.6 / 7	3.6 / 7
5	Fan Diameter & No. of Fans / Ph	300/1/1ph	300/2/1ph	300/3/1ph	350/3/1ph	350/3/1ph	450/2/3ph	450/3/3ph
6	Air Flow (m^3/h) / Throw(Mtr)	1560/9	3120/9	4680/10	8200/12	8200/12	8600/14	12900/14
7	Motor - Input Value (watts)	87 x 1	87 x 2	87 x 3	150 x 3	150 x 3	320 x 2	320 x 3
8	Motor - Current (Amp.)	0.42 x 1	0.42 x 2	0.42 x 3	0.65 x 3	0.65 x 3	0.74 x 2	0.74 x 3
9	Pipe Inlet & Outlet	1/2 & 7/8	1/2 & 7/8	1/2 & 7/8	5/8 & 1.7/8	5/8 & 1.1/8	5/8 & 1.1/8	5/8 & 1.3/8

EVAPORATOR UNIT (Evap. Temp. 10°C to -4°C)						
Sr. NO	Parameters/Model	CLS709-252A	CLD713-252A	CLD719-252A	CLD724-253A	CLD729-253A
1	Capacity (kW) (at -2°C, DT1= 8 K)	1.45	2.35	3.2	5.2	6.1
2	Heat Transfer Area (Sq. Mtr)	9.76	13.02	19.52	24.4	29.3
3	Tube Volume (In Ltr)	1.92	2.56	3.84	4.7	5.64
4	Fin Spacing-mm/FPI(Fin Per Inch)	3.6/7	3.6/7	3.6/7	3.6/7	3.6/7
5	Fan Diameter & No. of Fans	250/2	250/2	250/2	250/3	250/3
6	Air Flow (m^3/h) / Throw(Mtr)	2000/4	2000/4	2000/4	3000/4	3000/4
7	Motor - Input Value (watts)	50 x 2	50 x 2	50 x 2	50 x 3	50 x 3
8	Motor - Current (Amp.)	0.25 x 2	0.25 x 2	0.25 x 2	0.25 x 3	0.25 x 3
9	Pipe Inlet & Outlet	3/8 & 1/2	3/8 & 1/2	3/8 & 5/8	3/8 & 3/4	3/8 & 3/4

Air flow (m^3/h) :- The air flow is determined on a suction side chamber testing stand according to ISO 5801 with dry cooler surface.

Air throw(m) :- The air throw gives the distance from the outlet area of the air cooler at which the average of the air velocity taken at 0.5 m, 0.75 m and 1 m from the ceiling at 20°C equals 0.5 m/s.

Capacity (kW) :- The capacity data are based upon measurement according to standard at the following conditions :

- Refrigerant R404A,
- Liquid temperature 30°C resp. 20°C (for evaporating temperatures below -20°C)
- The superheat of refrigerant between outlet and inlet is considered 65%

The selection diagram and the capacity table are already considering the influence of the air humidity and specify the actual capacity of the cooler under operating conditions (wet or frosty cooler surface).

RIPENING CHAMBER

Ice Make Refrigeration Limited provides most advanced fruits ripening rooms for fruits like mango, banana and papaya. We provide forced draft cooling system for uniform ripening of fruits even in large capacity rooms. We use blended gas system for the purpose of ripening which is acceptable as per universal standard. With our system, client has unique advantage of controlling ripening cycle from four to fifteen days! We provide precise gas monitoring system for better storage life of the produce.





The essential requirements of an ethylene ripening system are:

• A reasonably air tight room with insulation • A temperature control system for cooling and/or heating
• An air circulation and ventilation system • Humidity control • An ethylene gas injection system
• An electric control system

We Offer Ripening Chamber

Manual (Time Based), Semi Automatic & Automatic
We ensure that each ripening chamber is designed to maintain the precise conditions required for specific application. Features of our ripening chamber such as,

- Microprocessor controlled and compatible with computer
- Low Cost and light weight, constructed with PUF-panels
- Low investment and maintenance cost
- Easy to commission, Swing or Sliding door
- Ripening capacities from 5 tons to 25 tons for each chamber
- Moveable from one site to another and expandable
- Controlled levels of Ethylene, CO^2, Temperature, humidity & time
- Energy efficient, uniform and quality ripening
- Ripening chambers are reliable with energy-efficient refrigeration units

Gas Emission Systems Features

- Safe
- Flexible system
- Intelligent control
- Independent settings
- Programmable

Daily Ripening Chart

	Days 1	Days 2	Days 3	Days 4	Days 5	Days 6	Days 7	Days 8
8 Days	14.5°	14.5°	14.5°	14.5°	14.5°	14.5°	14.5°	14.5°
7 Days	15.5°	15.5°	15.5°	15.5°	14.5°	14.5°	14.5°	14.5°
6 Days	16.5°	16.5°	15.5°	15.5°	14.5°	14.5°	14.5°	
5 Days	16.5°	16.5°	16.5°	16.5°	15.5°	14.5°		
4 Days	18.0°	18.0°	16.5°	15.5°	14.5°			



1	2	3	4	5	6	7
Green Natural Green	Light Green First Change in colour as a result of ripening	Light Green with Light Yellow Clear Change in colour ready for market in hot weather	Yellow with some green Ready for market in cold weather	Yellow with Green at ends Ideal colour for retail sale	Full Yellow Ready for sale and for eating	Yellow with Brown spots Fully ripened with aroma

Ethylene Exposure:

Sr.No.	Product Details	Ethylene Concentration (PPM)	Ethylene Exposure Time (Hours)	Ripening Temperature (°C)	Storage Temperature after ripening(°C)
1	Banana	100-150	24-48	15-18	13-14
2	Mango	100	24	20-22	10-13
3	Papaya	100	24-48	20-25	About at 7

OTHER BUSINESS VERTICALS

Industrial Refrigeration

- Water Chilling Plant
- Brine Chilling Plant
- Oil Chilling Plant
- Air Chilling Plant
- Ice Building Tank (IBT)

Ammonia Refrigeration

- Turnkey Projects for Cold Stores
- Water Chillers for Dairy
- Glycol Chillers for milk deep chilling for dairy
- Glycol chillers for beverage plants
- Glycol chillers for brewery plants
- Water Chillers for Pharma
- LP Receivers with Ammonia Pumps for retrofit plants
- Ice Accumulating Coils
- Atmospheric Condensers

Commercial Refrigeration

- Ice Cream Mix Plant
- Batch Pasteurizer
- Ageing Vat
- Bulk Milk Chiller
- Ice Candy Production Machine
- Ice Cream Hardener (Static/Tunnel Type)
- Mini Blast Freezer & Chiller
- Heat Pump Food Dehydrator

Transport Refrigeration

- Refrigerated Vehicle (GRP)
- Refrigerated Van - Eutectic (GRP)
- Bunk House
- Dry Insulated Container (GRP)

OUR VALUED CLIENTS

NETWORK

- Head Office & Maufacturing Plant-I , DantaliGandhinagar, Gujarat
- Manufacturing Plant-II, Vamaj, Gujarat
- Manufacturing Plant-III, Chennai, Tamilnadu
- Branch Offices, Dealers & Associates and Company Representatives



PROJECT DETAILS

Vadera Trade Link Pvt. Ltd.
Product : Multi Commodity
Product Range : Multi Storey Cold Storage
Storage Capacity Approx.: 4000 Metric Ton
Location : Barmer, Rajasthan

Bombaywalla Puranpoli Pvt. Ltd.
Product : Ready to Eat Frozen Food
Product Range : Blast Chiller, Blast Freezer & Cold Storage.
Storage Capacity Approx.: 700 Metric Ton
Location : Ahmedabad

Ariba Food Products Pvt. Ltd.
Product : Ready to Eat Frozen Food
Product Range : Blast Chiller, Blast Freezer & Cold Storage.
Storage Capacity Approx.: 185 Metric Ton
Location : Indore

Kitchen Express Ltd.
Product : Spices & Grocery
Product Range : Cold Storage
Storage Capacity Approx.: 2500 Metric Ton
Location : Ahmedabad

Himalayan Food Park
Product : Tomato
Product Range : Cold Storage
Storage Capacity Approx.: 1250 Metric Ton
Location : Uttarakhand

Hansaraj Exports
Application : Fruits Storage
Product Range : Cold Storage
Storage Capacity Approx. : 1500 Metric Ton
Location : Jammu

Prestige Ice Cream
Product : Ice Cream
Product Range : Cold Storage
Storage Capacity Approx. : 4 Lakhs Litres
Location : Kolkata

M.A. Banana Supplier
Product : Banana
Product Range : Banana Ripening - Multi Chamber
Storage Capacity Approx. : 88 Metric Ton
Location : Baroda

India Dairy Product Ltd.
Product : Ice Cream & Milk
Product Range : Cold Storage
Storage Capacity Approx. : 56,000 CFT
Location : Kolkata

N.B. Ice Cream
Product : Ice Cream
Product Range : Cold Storage
Storage Capacity Approx. : 3.75 Lakhs Litres
Location : Ranchi

Swaraj India Industries Ltd.
Product : Butter, Dairy Product like Butter & Ghee
Product Range : Blast Freezer and Cold Storage
Storage Capacity : 130 MT
Location : Satara, Maharashtra

Fortune Dairy Industries Pvt. Ltd.
Product : Butter, Dairy Products Like Ice cream, Butter, Milk, Curd
Product Range : Blast Freezer, Blast Chillers & Cold Storage
Storage Capacity : 100 MT
Location : Pune, Maharashtra

Best Foods Marketing Pvt. Ltd.
Product : Dates
Product Range : Cold Storage
Storage Capacity : 1550 MT
Location : Colombo, Sri Lanka

Pawanshree Foods International Pvt. Ltd.
Product : White Butter
Product Range : Cold Storage
Storage Capacity : 400 MT
Location : Maksi, Madhya Pradesh

Sri BalaSanka Mill
Product : Dried Pulses, Tamarind
Product Range : Low Humidity Cold Storage
Storage Capacity : 250 MT
Location : Theni, Tamilnadu
Executed by Our Sister concern : Bharat Refrigeration Pvt. Ltd.

ICE MAKE
Trusted Cooling Partner

WEBSITE CATALOGUE FEEDBACK

ICE MAKE REFRIGERATION LIMITED

Commercial & Industrial Refrigeration Equipment Manufacturer & Exporter

An ISO 9001:2015, ISO 14001:2015 and ISO 45001:2018 certified Company

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